FM exam scope update

14 October 2026

Prioritise the material that will be tested. The professor's clarification applies to the half course exam and the Part I questions of the full exam.

Excluded from the exam

- Money market hedge, page 88.
- Currency swaps, pages 109 to 117.
- Discounted Cash Flow valuation.
- Guest speaker lectures.

How to use this fiche

The original source explanations are retained so the course record remains complete. Do not treat the excluded sections as required exam preparation.

Session 9: the LBO and multiples material is not named as excluded. The DCF valuation method is excluded, not the entire LBO session. The appendix at the end of this fiche explains the new revised class workbook and reconciles its sponsor return.

Source: Blackboard announcement "Exam content clarifications", published 13 October 2026 at 19: 06 Europe/Rome.

Leveraged Buyouts & Value Creation (Session 9)

By the end of this session you must know what private equity is and how LP/GP funds are structured and paid, the three engines of value creation (A-L-O), what an LBO is and what makes a good target, why leverage amplifies the equity return, who the players in a deal are and the four exit routes, how to build an LBO model in eight steps (entry valuation, sources & uses, debt tranches, cash sweep, returns), how to choose and apply multiples, and how to test whether a structure is safe.

Plan

1. Private equity and how PE funds work
2. The three engines of value creation (A-L-O)
3. What an LBO is, and what makes a good candidate
4. Why leverage amplifies the equity return
5. Deal structure: the players and the four exits
6. The LBO model in eight steps
7. Choosing and applying multiples
8. Returns: IRR, MOIC and the five levers
9. Is the structure safe? Credit statistics
10. Common Mistakes & Exam Traps
11. Must Know Absolutely

1. Private equity and how PE funds work

Private equity is equity capital invested in companies not listed on a public exchange, to create value over a multi-year hold and realise a gain at exit.

Defining features:

- **Not public** — privately held shares (though a listed firm can be acquired and taken private).
- **Illiquid** — no daily price; you cannot trade in and out as with a public stock.
- **Exit value is what matters** — IRR and CoC (Cash on Cash, also MoM — Money on Money) at exit are the key metrics, encouraging genuinely long-term decisions.
- **Controlling ownership** — PE typically holds control, unlike a public company's dispersed shareholders.

How PE managers create value:

- **Buy-and-build** — acquire growth companies in fragmented industries (clinics, vets, funeral homes).
- **Stage capital** — invest at critical inflection points of development.
- **Turn around** — restructure, refocus or revitalise inefficient operating companies.

The limited-partnership structure:

- **Limited Partners (LP)** commit capital: pension funds, endowments, insurance firms, family offices, funds of funds, sovereign wealth funds, high-net-worth individuals. Minimum tickets depend on fund size (generally > $1-10m); fund lifetime is contractually fixed (10-12 years).
- **General Partners (GP)** select, invest and manage: pooled venture capital funds, buyout funds, mezzanine debt funds.
- **GP fees** — management fees typically 1-2%, plus **carried interest** of 20% subject to a hurdle (earned only if the fund achieves the minimum required return for LPs). Investment period: 3-5

years (up to 7).

The right fund depends on the life-cycle phase (same map as Session 8): seed/Series A — business angels and VC; Series B/C — venture and development capital; maturity — bank financing and PE/LBOs; decline — turnaround funds. Founder ownership decreases, external and financial ownership increases, then IPO or M&A exit.

> **Core message:** the buyout window — LBO funds target **mature** businesses whose stable cash flows can support acquisition debt.

2. The three engines of value creation (A-L-O)

The fundamental drivers of a sponsor's IRR:

- **A — Arbitrage (multiple expansion).** Buy at a relatively low price and exit substantially higher — more art than science. Proprietary advantage lets PE buy below market; public-private arbitrage (buy private, sell public); the exit multiple expands with a synergistic strategic buyer.
- **L — Leverage (debt paydown).** Use the target's free cash flow to repay the acquisition debt over the hold. Tax benefit: interest is deductible (the shield). Governance benefit: debt imposes discipline on management.
- **O — Operational (margin improvement).** Lift EBIT/EBITDA margins via reorganisation and efficiency: grow the top line (volume, price, new markets), cut COGS and SG&A, accelerate the cash-conversion cycle.

> **Example (Carlyle & Moncler):** entry at 11.8× EBITDA; the value-creation plan focused on organisation, geographical expansion, insourcing of key activities, supply chain, diversification and best practices for exit. (The companion file [for class] LBO_example.xlsx quantifies the three engines.)

3. What an LBO is, and what makes a good candidate

An **LBO** is the acquisition of a company financed mostly with debt and only a thin slice of equity. A financial sponsor puts up the equity; banks and bond investors provide the rest.

- **The target** — a public company taken private, a corporate carve-out, or a secondary buyout from another sponsor.
- **The engine** — the acquired company's own cash flow services and repays the acquisition debt over the hold.
- **The objective** — sell in ~5 years at a higher EBITDA and/or multiple, having de-levered along the way.
- **The by-product** — it sets a "floor" value: the most a sponsor can pay and still clear its hurdle rate.
- **Funding mix** — debt ~60-80%, equity ~20-40%. Purchase price = Debt + Equity.

What sponsors hunt for (ten traits of a good candidate):

1. **Undervalued vs. peers** — trades at a discount to comparable companies.
2. **Underperforming vs. peers** — margins or growth lag, leaving room to fix operations.
3. **Awaiting a growth catalyst** — needs capital or a new owner to trigger expansion.
4. **Under-leveraged** — little debt relative to its cash flow.
5. **High cash balances** — excess cash can help fund the purchase.

6. **High barriers to entry** — sustainable advantages protect margins.
7. **Stable operating cash flows** — predictable cash to service interest through the cycle.
8. **Monetisable assets** — non-core assets can be sold to repay debt.
9. **Moderate capex & R&D** — more free cash flow left for debt paydown.
10. **Strong management** — able to execute the value-creation plan.

4. Why leverage amplifies the equity return

Same company, bought for €100 and sold for €180 after five years; holding the debt flat, the more of the purchase funded with debt, the larger the return on the shrinking equity cheque.

	Equity in	Equity out	MOIC
All-equity	€100	€180	1.8×
50% debt	€50	€130	2.6×
70% debt	€30	€110	3.7×

The mechanism:

- As the target's cash flow pays down debt, enterprise value is steadily **converted into equity value**.
- A smaller equity base at entry means each euro of value creation lands on fewer shares — **magnifying the return (and, symmetrically, the downside)**.

5. Deal structure: the players and the four exits

The cast:

- **Financial sponsor** — the PE fund; supplies equity, controls the deal.
- **Lenders** — banks and bond investors; supply the debt tranches.
- **NewCo / BidCo** — the acquisition vehicle that borrows, buys the target, then merges with it.
- **Management** — often rolls over equity and runs the value-creation plan.
- The target's own cash flow services and repays the acquisition debt.

Four exit routes after a ~5-year hold:

1. **Strategic sale** — sell to a corporate (trade) buyer; often the highest price thanks to synergies.
2. **Secondary buyout** — sell to another sponsor that sees further value to create.
3. **Initial public offering** — list the company and sell down over time.
4. **Dividend recapitalisation** — re-lever to pay a dividend; returns cash without a full exit.

6. The LBO model in eight steps

A disciplined sequence — each step feeds the next, ending in the sponsor's return on equity: (1) size the deal, (2) sources & uses, (3) set the debt tranches, (4) project the P&L, (5) derive free cash flow for principal repayment, (6) build the debt schedule, (7) link it up (circularity), (8) compute the returns.

Step 1 — Size the deal (entry EV and equity value).

- Public target: equity value = number of diluted shares × price per share.
- Private target: start from a normalised, last-twelve-months **EBITDA**; apply an **entry multiple** benchmarked to comparable transactions and trading peers: Entry EV = entry multiple × EBITDA; bridge to equity: Equity value = EV − net debt assumed; add transaction fees and a minimum cash buffer to get total uses.
- Illustrative deal: EBITDA (2026) $409.0k × 8.5× = entry EV $3,476.5k (debt-free/cash-free, so entry equity value = $3,476.5k).

> **Core message:** the price paid is the single biggest determinant of the return. Paying one turn of EBITDA less at entry lifts the IRR in every exit scenario — disciplined sponsors walk away from over-priced auctions rather than rely on multiple expansion.

Step 2 — Sources & uses. The two columns must balance: every dollar raised funds a dollar of the transaction.

- Uses: purchase price $3,476.5k + transaction fees $52.1k (1.5% of TEV; fees are typically 1-2% of TEV) = total uses $3,528.6k. (Plus: refinance net existing debt if a change-of-control clause is triggered, cash for operations.)
- Sources: Term Loan B 3.2× EBITDA at SOFR+2.25% = $1,308.8k (37.1%); senior notes 1.5× at 10.0% = $613.5k (17.4%); subordinated notes 1.0× at 12.0% = $409.0k (11.6%); total debt 5.7× EBITDA = $2,331.3k (66.1%); sponsor equity $1,197.3k (33.9%); plus an undrawn revolver.
- **Balancing item: sponsor equity is the plug** — it absorbs whatever the debt tranches and minimum cash leave unfunded, so total sources always equal total uses.

Step 3 — Set the tranches (the LBO debt stack). Layer from cheapest/most senior at the top to equity at the bottom, sized in turns of EBITDA:

Layer	Pricing	Features
Revolver (RCF)	SOFR + 200 bps	Senior; generally not drawn at completion, reserved for working capital; ~5 years
Term Loan A (amortising)	SOFR + 300 bps	Senior secured, ~2.0×, banks (pro-rata), ~5 years
Term Loan B (bullet)	SOFR + 350 bps	Senior secured, ~2.5×, institutional, SOFR floor 0.75-1.00%, ~7 years
Senior unsecured notes	~7.0%	~8-year maturity, callable after 4 years at par + ½ coupon
Subordinated notes	~8.5%	1-2 years longer than senior bonds, call at par + ½ coupon
Sponsor equity	the residual cheque	Junior, first-loss

- Total leverage often ~3-6× EBITDA, set by market appetite.
- Pricing steps up down the stack: each bond layer adds ~100 bps for weaker security/seniority and ~50 bps for longer maturity.
- Floating tranches price over **SOFR**, which replaced USD LIBOR in 2023.

Steps 4-5 — From P&L to free cash flow.

- Step 4: project the income statement from the last 3-5 years of financials plus management forecasts, analyst projections and own analysis (inputs: revenue growth, gross margin, SG&A).
- Step 5: levered FCF for principal repayment = EBITDA − maintenance capex (total capex less growth capex, often % of revenue) − change in net working capital (often grown in line with revenue) − cash interest − cash taxes.

Step 6 — Debt schedule and the cash sweep. Where free cash flow goes, in contractual priority order:

1. **Mandatory amortisation** — TLA ~20%/yr and TLB ~1%/yr are repaid first, on schedule.
2. **Cash sweep → Term Loan A** — excess FCF prepays the TLA until it is gone.
3. **Cash sweep → Term Loan B** — then the sweep turns to the TLB.
4. **Bonds and cash balance** — notes repay at maturity (bullets); any residual builds cash.

- The cash sweep is **forced de-levering**: loan terms compel the company to apply surplus cash to prepay term loans.
- Notes cannot be swept (high-yield bonds may be called early, at a premium); the order follows seniority.
- In the illustrative deal: total debt falls from $2,331.3k to $1,668.8k over five years — from 5.7× to 3.7× EBITDA. Only the TLB moves (1,308.8 → 646.3, i.e. 51% repaid): 1% mandatory amortisation ($13.1k/yr) plus the sweep of all excess FCF ($597.0k over five years). TLA and revolver stay undrawn; senior ($613.5k) and subordinated ($409.0k) notes are bullets.

Step 7 — Link it up (circularity). Interest expense depends on debt balances, which depend on the cash sweep, which depends on free cash flow, which depends on interest expense — the model iterates until it converges.

Step 8 — Compute the returns (next section).

7. Choosing and applying multiples

The fundamental logic: Multiple = price you pay for the asset / profit generated by the asset. The numerator can be **equity value** (price of the equity only — pairs with equity-side profits) or **enterprise value** (equity value + market value of debt — pairs with asset-side profits). The denominator can be revenue, profit or cash flow.

- **EV/EBITDA** (most used in LBO): independent of financial leverage and capital structure; enables comparability; reliable for cyclical industries. Distorted by differences in capex, tax policies and leverage.
- **P/E**: widely used (especially forward P/E); EPS estimates available; but affected by leverage and accounting standards.
- **EV/EBIT**: independent of leverage; suitable for cyclical companies; distorted by accounting-policy differences.
- **Price/Book**: useful for capital-intensive businesses and financial institutions; sensitive to accounting differences; less relevant for growth companies.
- **EV/Sales**: useful with negative earnings or early-stage companies; less accounting-sensitive; but only a partial picture.

> **Core message:** no single multiple is the right one. EV-based multiples neutralise capital structure; equity-based multiples capture the levered return but absorb leverage and accounting choices. Cross-check at least two and explain the gap.

Let the sector economics pick the denominator: cyclical — EV/EBITDA; manufacturing — P/E, EV/EBIT; start-up — EV/Sales (revenue multiples are often the only tool when EBITDA or net income are negative); infrastructure — EV/EBITDA (high fixed costs and depreciation, stable cash flows); financial services — Price/Book (book value reflects real valuation); retail — EV/Sales and P/E with normalised earnings (margins converge over time).

Consistency rule (P/E example): a P/E is comparable only if price and EPS are defined the same way across the peer set — price (current vs 6-12 month average), EPS (basic vs diluted; before/after extraordinary items), timing (actual vs trailing TTM/LTM vs forward). **State the definition before you compare.**

Building the comparable set: build the peer cluster on **fundamentals, not on sector alone** — only peers with similar growth and margins are true comparables.

> **Example (Bulgari-LVMH, trading multiples):** screening fifteen luxury names on category, growth and margins keeps only the true peers (Hermès sets the top of the range; mid-market, accessible-luxury, eyewear and apparel names are dropped). Applying peer multiples to Bulgari's 2011 figures gives €5.6-13.7 a share; relevant comps give €7.3-13.7 against €5.6-9.7 on full-sample averages; averaging all six methods gives an equity value of €2,676m (€8.89/share). **The peer set moves the price more than the method.** Mechanics: EV = base × multiple; equity value = EV − net financial position; P/E applies directly to net income (no NFP adjustment).

> **Example (Amazon / Whole Foods, transaction multiples):** precedents are filtered on sector fit, strategic bidder, 100% control, cash consideration and recency — one decisive failure disqualifies a deal (e.g. Apollo/Fresh Market dropped as a financial sponsor; Kraft-Heinz dropped as a stock mega-merger; Sun Art dropped as a minority stake). Median of the four retained deals: EV/EBITDA 15.0×, EV/EBIT 24.5×. Applied to Whole Foods: implied €53.84/share (EV/EBITDA) and €49.55 (EV/EBIT) versus the €42 offer... check the slides for the final verdict. The median selection smooths outlier impact.

8. Returns: IRR, MOIC and the five levers

- **IRR (internal rate of return)** — annualised; captures both the size and the timing of cash flows.
- **MOIC (cash-on-cash)** — cash in ÷ cash out; ignores timing.
- Both are reported together: IRR for timing, MOIC for absolute scale.

Base case of the illustrative deal: **~12.1% IRR / 1.8× MOIC** (enter and exit at 8.5×, five-year hold to 2030). Exiting a year early cuts IRR to 9.6%; exiting 1.0× lower (7.5×) cuts it to 6.9%; at 8.0× or below, any exit before 2028 loses money.

The five levers that move the return:

1. **Purchase price** (A) — pay less: lower entry, higher return.
2. **Exit multiple** (A) — sell higher: multiple expansion lifts proceeds.
3. **Leverage** (L) — more debt: smaller equity cheque, but more risk.
4. **Revenue growth** (O) — grow faster: bigger EBITDA base at exit.
5. **EBITDA margin** (O) — run leaner: more cash from every euro of sales.

> **Core message:** leverage is the lever that cuts both ways — it raises return and risk together. The other four improve the deal's quality; leverage only changes how the same outcome is split

between debt and equity, amplifying gains and losses alike.

9. Is the structure safe? Credit statistics

Three families of ratios test the structure each year — and a healthy LBO gets safer as it ages:

- **Leverage** — Net Debt/EBITDA and Senior Secured/EBITDA: is debt too large a share of the business?
- **Coverage** — EBITDA/interest and (EBITDA − capex)/interest: can the business comfortably pay its way?
- **Liquidity** — cash on hand + undrawn revolver: is there a buffer against near-term needs?

Base case of the illustrative deal: Net debt/Adj. EBITDA falls from 5.56× (2026) to 3.74× (2030) — inside the ≤ 6.0× safe zone; Senior secured/Adj. EBITDA from 3.06× to 1.45× (safe zone ≤ 4.0×).

10. Common Mistakes & Exam Traps

- Confusing CoC/MoM (cash multiple, no timing) with IRR (annualised, timing matters) — report and interpret both.
- Believing leverage creates value by itself: it amplifies an outcome; only A (price) and O (operations) create value.
- Forgetting that sponsor equity is the **plug** in sources & uses — it absorbs whatever debt does not fund.
- Sweeping bonds: the cash sweep prepays **term loans** only; notes are bullets repaid at maturity (callable at a premium).
- Comparing multiples with inconsistent definitions (diluted vs basic EPS, trailing vs forward, extraordinary items) — state the definition first.
- Building a peer set on sector labels instead of fundamentals (growth and margins) — the peer set moves the price more than the method.
- Placing LBOs outside the maturity phase of the life cycle, or targeting firms with volatile cash flows and heavy capex.
- Reading total leverage only: senior secured leverage and coverage ratios tell you whether the structure is actually safe.

11. Must Know Absolutely

Must know:
• PE = illiquid, controlling equity in unlisted companies; LP commits capital, GP manages for 1-2% fees + 20% carry over a hurdle; funds live 10-12 years.
• Value creation = A-L-O: arbitrage (multiple), leverage (paydown), operational (margins).
• LBO = ~60-80% debt / 20-40% equity; the target's own cash flow repays the debt; target profile: undervalued, underperforming, under-leveraged, stable cash flows, moderate capex, strong management.
• Leverage amplifies MOIC because debt paydown converts EV into equity value on a shrinking equity base — symmetrically for losses.
• The 8-step model: size the deal (EBITDA × entry multiple) → sources & uses (equity is the plug) → tranches (RCF → TLA → TLB → senior → subordinated → equity) → P&L → FCF → debt schedule

(mandatory 1% TLB + cash sweep; bonds are bullets) → circularity → IRR/MOIC.
• Cross-check at least two multiples with consistent definitions; peer set on fundamentals; EV = base × multiple, equity = EV − NFP.
• Five return levers: purchase price, exit multiple, leverage, revenue growth, EBITDA margin.
• Safety = leverage (≤ ~6× net debt/EBITDA), coverage (EBITDA/interest), liquidity (cash + revolver); a healthy LBO de-levers as it ages.

Watch Out

Watch out: the detailed P&L projection (Step 4) and some tables (Carlyle/Moncler figures, Whole Foods final verdict line) are partly graphical in the slides — verify numbers on the original PDF and in [for class] LBO_example.xlsx. The HBS note "Note on LBO Capital Structure" (Ivashina) is optional reading for sessions 07-08-09.

Session 9: revised LBO class workbook

Updated 14 October 2026. The revised workbook replaces [for class] LBO_example.xlsx. Its figures below are separate from the illustrative debt stack in the original fiche.

8.1 Revised class workbook: follow one deal from entry to exit

The new workbook has an IRR and MoM sheet and a Key Steps sheet. This is a separate example from the detailed debt stack above. Do not mix its numbers with that example.

1. **Price the business.** Entry EBITDA is 100 and the entry multiple is 10. Enterprise value is 100 × 10 = **1,000**. Existing net debt is 30, so the seller's equity proceeds are 1,000 minus 30 = **970**.
2. **Fund the transaction.** Refinance existing net debt 30 and pay fees of 2% × 1,000 =20. Total uses are 970 +30 +20 = **1,020**. New debt is 4 × EBITDA 100 = **400**. Sponsor equity is the plug: 1,020 minus 400 = **620**. Uses and sources both total 1,020.
3. **Project operating cash flow.** Revenue starts at 250, grows 10% annually and earns a 40% EBITDA margin. Exit EBITDA in year 5 is 250 × 1.1^5 × 40% = **161.051**. Cash interest uses beginning of period debt at 10%, not average or ending debt. This prevents circular references in this workbook.
4. **Repay debt.** Tax is 40% of pretax income, capex is 4.5% of revenue and extra working capital is 15% of the revenue increase. Cash available for repayment is EBITDA minus taxes, working capital investment, capex and cash interest. Repayment is the smaller of that cash flow and beginning debt, so debt cannot become negative. Debt falls from 400 to **173.322127** by exit. The workbook assumes no cash build up during repayment.
5. **Bridge exit EV to equity.** An 11× exit multiple gives 161.051 ×11 = **1,771.561** enterprise value. Deduct exit net debt 173.322127 to get sponsor proceeds **1,598.238873**.
6. **Measure the sponsor return.** MoM = 1,598.238873 /620 = **2.577805×**. With only the initial outflow and the year 5 inflow, IRR = (2.577805)^(1/5) minus 1 = **20.8509%**. For interim distributions, use the full cash flow IRR instead of this shortcut.

Explain where the gain came from. Sponsor proceeds minus the initial cheque = 1,598.238873 minus 620 = **978.238873**. The attribution must reconcile to that number.

Driver	Calculation	Gain
EBITDA growth at the entry multiple	(161.051 minus 100) ×10	610.510000
Multiple expansion on exit EBITDA	(11 minus 10) ×161.051	161.051000
Acquisition debt repaid	400 minus 173.322127	226.677873
Transaction fees	Deduct the entry fee	(20.000000)
Total sponsor gain	Sum of the four rows	978.238873

Sensitivity, not a promise. In the workbook, 4× entry leverage gives IRRs of 15.5317%, 18.3108%, 20.8509% and 23.1939% at exit multiples 9×, 10×, 11× and 12×. At the 11× exit multiple, 3× leverage gives 19.1917% and 7× gives 30.1884%. Higher leverage increases the model's equity return but also its financing and downside risk. Do not call the sensitivity table a forecast.

Source: (for class) LBO_example_OP_rev RA.xlsx, sheets IRR and MoM and Key Steps.